SYSTEM ARCHITECTURE & PRODUCT DESIGN SPECIFICATION

Project Nexus-PC (Stratum Engine v3.0)

Universal SoC Abstraction & Multi-Tier Snapdragon Execution Platform

Project Codename: Nexus-PC
Lead Engineering: Flutter Team
Target Architecture: Universal Snapdragon 8 Series (Gen 2 to Elite Gen 6+)
Spec Version: 3.0.0 (Dynamic Alias Normalized)
Target OS: Android 12+ (API 31+)
Date: October 2026

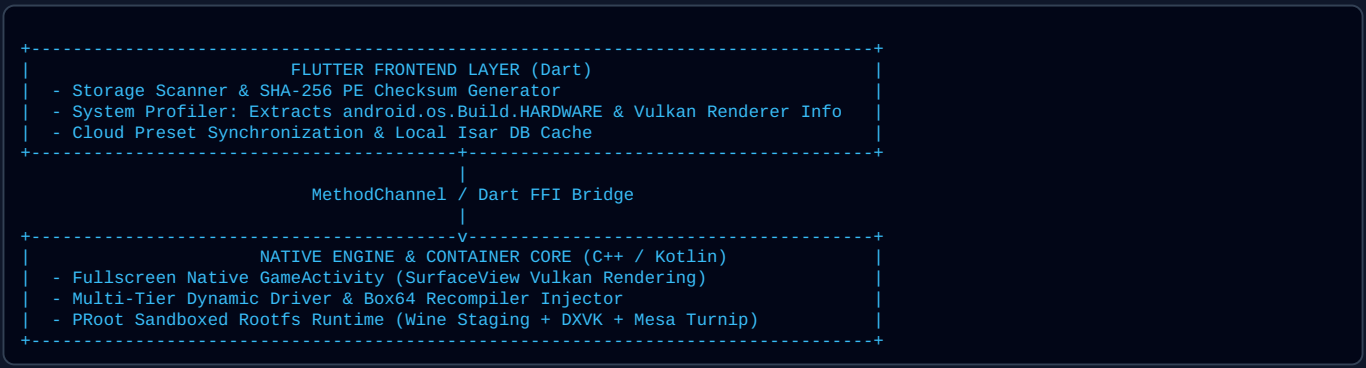
1. Executive Summary & Vision Statement

Project Nexus-PC provides a zero-configuration, plug-and-play PC game runner for Android devices. To overcome hardware market volatility and Qualcomm's unpredictable branding shifts (ranging from standard numbering to *Elite*, *Gen 5*, *Elite Gen 6*, and *Extreme Gen 6*), version 3.0 introduces a robust **SoC Alias Normalization & Capability Tiers** architecture.

By decoupling hardware naming conventions from execution profiles through a cloud-synced backend normalization engine, the application ensures complete future-proofing. New mobile chipsets automatically integrate without requiring client application updates on the Google Play Store.

> **Core Design Postulate**
> *"Zero Configuration & Future-Proof Abstraction."* Raw device hardware strings are dynamically normalized on backend servers, mapping unfamiliar chip monikers into standardized performance capability tiers instantly.

2. End-to-End System Architecture



3. SoC Alias Normalization & Capability Tiers

Qualcomm's frequent marketing shifts create fragmentation risks. Nexus-PC eliminates this vulnerability via a three-stage server-side pipeline:

- **Raw Data Ingestion:** On startup, the Flutter client reads system properties (`android.os.Build.HARDWARE`, `BOARD`) and Vulkan/OpenGL renderer strings.
- **Backend Alias Map:** Unpredictable strings (e.g., ***snapdragon_8_elite_extreme_gen_6***) are passed to a dynamic server dictionary that normalizes the input to canonical internal system keys.
- **Capability Tier Grouping:** Instead of coding per-chip exceptions, chips are grouped into functional performance tiers governing compiler flags and driver packages.

Capability Tier	Representative Marketing Names	Target GPU Architecture	Optimized Execution Profile
Tier 3 (Ultra / 2nm Class)	Snapdragon 8 Elite Gen 6, Extreme Gen 6	Adreno 845+ / Next-Gen	Max-performance Box64 flags, aggressive clock profiles, cutting-edge Turnip builds for heavy DX11/12.
Tier 2 (High-End Flagship)	Snapdragon 8 Elite, Gen 5, Gen 3	Adreno 8-Series / 750	Balanced thermal profiles, stable 60 FPS across demanding 3D PC game libraries.
Tier 1 (Base Flagship)	Snapdragon 8 Gen 2 / 8+ Gen 1	Adreno 740 / 730	Stable compatibility baseline ensuring flawless execution of indie and classic 3D titles.

4. Automated Cloud Fingerprinting & Dynamic Adaptation

When a user launches a game, the client issues a normalized API query:
GET /api/v1/profile?hash={GAME_HASH}&soc_alias={NORMALIZED_KEY}&ram={DEVICE_RAM}

- **Dynamic Driver Swapping:** The backend serves the precise Mesa Turnip driver package mapped to the resolved Adreno generation.
- **RAM-Aware Allocation:** The container inspects physical device RAM (12GB to 24GB+) to dynamically configure Box64 recompiler cache bounds and Wine memory limits, preventing Out-Of-Memory (OOM) crashes.

5. Project Cost & Implementation Roadmap

Category	Bootstrap Cost	Production Cost	Notes
Google Play Developer Account	$25.00 (One-time)	$25.00 (One-time)	Publishing license fee.
Cloud Backend & CDN	$0.00 (Supabase Free)	$25.00 / month	Lightweight normalized JSON profiles.
Test Hardware Infrastructure	$0.00 (Existing Device)	$350.00	Verification across Gen 2, Gen 3, and Elite platforms.
TOTAL UPFRONT	**~$25.00**	**~$400.00**	**Lean, scalable indie launch budget.**

Development Roadmap: Phase 1 (Flutter UI & Bridge, Weeks 1-4) → Phase 2 (Alias Normalization & Cloud Engine, Weeks 5-8) → Phase 3 (Zero-Latency Touch Overlay & Release, Weeks 9-12).